Team 26: Relationship Between Sleep, Study Habits, Lifestyle Factors, and Academic Performance

Analyzing how sleep duration, study habits, and daily lifestyle choices influence student GPA

1 1. Main Hypothesis / Question

Question

What is the relationship between sleep, study habits, lifestyle factors, and academic performance (GPA)? Specifically, how do sleep duration, study hours, screen time, and physical activity jointly predict student academic outcomes, and which lifestyle habits have the strongest impact on GPA?

Background

A student's academic success depends on a complex interplay of personal routines and lifestyle choices. While traditional focus has centered primarily on time spent studying, emerging evidence highlights the vital role of sleep quality, digital exposure, and physical well-being.

The **Student Lifestyle Dataset** encompasses a range of interrelated lifestyle dimensions, including daily screen time (hours), sleep duration (hours), study hours, physical activity level, and cumulative GPA. Rather than evaluating these variables in isolation, analyzing the combined influence of sleep duration, study habits, and lifestyle routines provides a more comprehensive model for understanding student performance.

Our Approach

We follow a structured data analysis pipeline on the dataset:

1. **Data Cleaning & Exploration**: Clean the dataset, handle missing values, and evaluate pairwise correlations among study hours, sleep duration, screen time, physical activity, and GPA.
2. **Statistical Modeling**: Build multiple regression models in Python to quantify the main and interactive effects of study habits, sleep, and lifestyle factors on GPA.
3. **Analysis of Relative Importance**: Assess the contribution and significance of each lifestyle and academic habit predictor on overall student academic outcomes.

Factor	What it measures	Example / Range
Academic Performance (GPA)	Student cumulative grade average	Range: 0.0 – 4.0
Study Hours	Time dedicated to academic study daily	1 to 8 hours/day
Sleep Hours	Total night-time rest duration	4 to 9 hours/day
Screen Time	Total daily time spent on digital devices	2 to 10+ hours/day
Physical Activity	Daily time engaged in exercise/movement	0 to 3 hours/day

Hypotheses

- H_0: Sleep duration, study habits, and lifestyle factors have no significant relationship with a student's academic performance (GPA).
- H_1: Daily study hours and sleep duration both exhibit a significant positive relationship with GPA.
- H_2: High daily screen time and low physical activity negatively impact student academic outcomes, even after controlling for study hours.
- H_3: There is a significant interaction effect between sleep duration and study hours, such that adequate sleep amplifies the positive impact of study time on GPA.
- H_4: A comprehensive regression model incorporating study habits, sleep duration, screen time, and physical activity predicts GPA significantly better than a model using study hours alone.

1.1 Source papers, blog, podcast

- **Kaggle Dataset**: Student Lifestyle Dataset
- **Research Reference**: Hale, L., & Guan, S. (2015). *Screen time and sleep among school-aged children and adolescents.* Sleep Medicine Reviews. https://doi.org/10.1016/j.smrv.2014.07.007
- **Blog Article**: *How Screen Time Impacts Sleep and Academic Performance* – Sleep Foundation. https://www.sleepfoundation.org/children-and-sleep/screentime-and-sleep-in-children

1.2 Programming languages / tools

Python (`pandas`, `NumPy`, `SciPy`, `statsmodels`, `scikit-learn`, `matplotlib`) for data analysis and regression models; GitHub for version control; Google Sheets for tracking.

1.3 Datasets

Primary Dataset: `student_lifestyle_dataset.csv` from Kaggle, containing student study hours, screen time, sleep hours, physical activity, and GPA.

1.4 Links to LLM chats

Gemini: https://share.gemini.google/QSQsM2ORA2x5

2 2. Logistics

2.1 Updated spreadsheet link

https://docs.google.com/spreadsheets/d/1RzafAVBAYhVqOZKkN-fq5Nu-PN3WOEgO/edit?usp=sharing&ouid=107962005788953761308&rtpof=true&sd=true

2.2 Breakup of work and how we plan to work

Work package	Work distribution	How we will implement	Output
Data Cleaning & EDA	Chetan	Clean dataset, handle missing values, analyze distribution of sleep, study habits, and lifestyle variables	Cleaned CSV, EDA Notebook
Statistical Modeling	Karthik, Mukesh	Build linear/multivariate regression models for sleep, study, and lifestyle predictors against GPA	Regression tables, plots
Analysis & Validation	Rishanth	Evaluate predictor importance, interaction terms, and model comparisons	Model metrics
Final Report & GitHub	All (Team 26)	Compile final report, slides, and GitHub repo	Final PDF, GitHub, slides

How we will work. We will coordinate our work through weekly milestones and clear communication. Daily updates will happen via our team group chat, and a shared Google Doc will be used for task tracking. We will attend Friday help sessions and hold a weekly team sync at the Dept. Library. Code will be shared on GitHub using feature branches.

3 3. Specific help needed from the teaching team

- Feedback on whether multivariate regression models and interaction terms (e.g., sleep duration × study hours) effectively capture the relationship between lifestyle choices and academic performance.
- Advice on dealing with potential multicollinearity among correlated lifestyle factors (e.g., study time vs. screen time vs. physical activity).